



2026.04.17

审核批准页

Review and approval

	姓名 Name	签名 Signature	日期 Date
起草人 (ASD) Author (ASD)	吴秋燕 Qiuyan Wu	吴秋燕	2026.04.17
审核人 (ASD) Reviewer (ASD)	费梦丹 Mengdan Fei	费梦丹	2026.04.17
审核人 (QA) Reviewer (QA)	王玲玲 Lingling Wang	刘曼	2026.04.17
批准人 (QA) Approver (QA)	刘曼 Man Liu	刘曼	2026.04.17

**标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程****Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method****1. 目的 PURPOSE**

建立 HLX49-Ab 分子排阻色谱法 (SEC) 标准操作规程, 以检测抗体中单体、高分子聚合物及低分子量物质含量, 确保检验人员的规范操作, 保证检验的正常进行。

Establish standard operating procedure (SOP) for HLX49-Ab Size Exclusion Chromatographic method (SEC) to determine the content of monomer, high molecular weight species and low molecular weight species of antibody to ensure standardized operation and normal running of the analysis.

2. 范围 SCOPE

本检验方法适用于测定本公司生产的 HLX49-Ab 抗体溶液、稳定性样品及该项目其他样品的 SEC 纯度和杂质。

This method is applicable to the determination of SEC purity and impurity for antibody solution, stability samples and other samples of HLX49-Ab manufactured by Henlius.

3. 定义 DEFINITIONS**3.1. HMWS—高分子聚合物**

HMWS—High Molecular Weight Species

3.2. LMWS—低分子量物质

LMWS—Low Molecular Weight Species

3.3. Monomer—单体

Monomer—Monomer

3.4. SEC—分子排阻色谱法

SEC—Size Exclusion Chromatography

3.5. d—天平实际分度值

d—Actual value of scale division

4. 职责 RESPONSIBILITIES**4.1. 检验人员应当严格根据本规程进行实验操作、记录、数据处理以及结果报告;**

Analysts should strictly follow this procedure for experimental operation, recording, data processing and result reporting.

4.2. 复核人员必须认真复核检验人员的记录填写及数据计算的准确性;

The reviewer should carefully review the accuracy of the record and data calculation.

4.3. ASD部门负责人或其授权人监督本SOP的执行。

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程
Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

The principal of ASD department or authorized designee is responsible for monitoring the implements of the SOP.

5. 参考资料 REFERENCES

5.1. BAD-GE-014 理化和E&L色谱柱使用标准操作规程

BAD-GE-014 The Operational Procedure of Chromatographic Column for Physichemical and E&L.

5.2. BAD-EQ-003 安捷伦高效液相色谱仪操作规程

BAD-EQ-003 Agilent High-Performance Liquid Chromatography Operating Procedure.

5.3. BAD-GE-053 ASD理化色谱峰的鉴别和积分处理步骤

BAD-GE-053 Chromatographic Peak Identification and Integration Procedures for ASD Physical-chemical Group.

6. 程序 PROCEDURE

6.1. 试剂与材料 Reagents and Materials

试剂材料名称 Reagents/Materials	级别/规格 Grade and Specification	厂家及货号 Manufacturer and Cat. No.
6.1.1. 二水合磷酸二氢钠 Sodium Dihydrogen Phosphate Dihydrate	AR 或以上 AR or higher	国药 (20040718) 或其他 Sinopharm (20040718) or others
6.1.2. 十二水合磷酸氢二钠 Disodium Hydrogen Phosphate Dodecahydrate	AR 或以上 AR or higher	国药 (100203673) 或其他 Sinopharm (100203673) or others
6.1.3. 氯化钠 Sodium Chloride	AR 或以上 AR or higher	国药 (10019318) 或其他 Sinopharm (10019318) or others
6.1.4. 氢氧化钠 Sodium Hydroxide	AR 或以上 AR or higher	国药 (10019718) 或其他 Sinopharm (10019718) or others
6.1.5. 甲醇 Methanol	HPLC	Tedia (MS1922-001) 或其他 Tedia (MS1922-001) or others
6.1.6. 样品瓶 Vial	2 mL	Agilent (5182-0714) 或其他 Agilent (5182-0714) or others
6.1.7. 样品瓶内插管 Vial Insert	400 μ L, 250 μ L	Agilent (5181-3377) 或其他 Agilent (5181-1270) 或其他 Agilent (5181-3377) or others

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程
Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

试剂材料名称 Reagents/Materials	级别/规格 Grade and Specification	厂家及货号 Manufacturer and Cat. No.
		Agilent (5181-1270) or others
6.1.8. 样品瓶盖 Vial Caps	9 mm	Agilent (5182-0717) 或其他 Agilent (5182-0717) or others
6.1.9. 水相滤膜 Aqueous Phase Filter	0.45 μm	Millipore (HVLP04700) 或其他 Millipore (HVLP04700) or others
6.1.10. 溶剂过滤器 Solvent Filter	N/A	Pall (2040) 或其他 Pall (2040) or others
6.1.11. 超纯水 Ultrapure Water	N/A	实验室超纯水系统自制或其他 laboratory-made by ultrapure water system or others
6.1.12. 分子排阻色谱柱 Nano BioCore SEC-300 Column	7.8 mm $\times$ 150 mm, 3 μm	Nano (B213-030030-07815S)
6.1.13. BioCore SEC-300保护柱芯 BioCore SEC-300 guard column	3 μm , 4.6 $\times$ 10 mm	B213-030030-04601S-B1

注 1: 相同功能的试剂和材料可相互使用。

Note 1: reagents and materials with the same function can be used with each other.

注2: 保护柱可以延长色谱柱的寿命, 且不影响检测结果, 可以根据情况选择接入保护柱。

Note 2: The guard column can extend the lifespan of the column without affecting the results, it can be used as needed.

6.2. 仪器与设备INSTRUMENTS AND EQUIPMENTS

仪器/设备名称 Instruments/Equipment	规格 Specification	厂家/型号 Manufacturer and Model
6.2.1. 高效液相色谱仪 High Performance Liquid Chromatography	HPLC	Agilent 1260
6.2.2. 移液器 Pipette	0.1-2.5 μL , 0.5-10 μL , 2-20 μL , 10-100 μL , 20-200 μL , 100-1000 μL , 0.5-5mL	Eppendorf 或其他 Eppendorf or others
6.2.3. 分析天平 Analytical Balance	d=0.01 g, d=0.0001 g,	Denver TP-202 或其他 Denver TP-202 or others Denver T-114 或其他 Denver T-114 or others

**标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程****Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method**

仪器/设备名称 Instruments/Equipment	规格 Specification	厂家/型号 Manufacturer and Model
6.2.4. 涡旋混合器 Vortex Mixer	N/A	其林贝尔 (QL-866) 或其他 Kylin-bell (QL-866) or others
6.2.5. 超纯水系统 Ultrapure Water System	N/A	Millipore Milli-Q advantage A10 system 或其他 Millipore Milli-Q advantage A10 system or others
6.2.6. 磁力搅拌器 Magnetic Stirrer	N/A	Lab Companion (HP-2000)或其他 Lab Companion (HP-2000) or others
6.2.7. pH计 pH Meter	精度 0.01 Precision to 0.01	Mettler Toledo S210 或其他 Mettler Toledo (S210) or others
6.2.8. 超声波清洗机 Ultrasonic cleaner	N/A	宁波新芝 (SB4200D) 或其他 Ningbo Xinzhi (SB4200D) or others

(注: 相同功能的仪器和设备可相互使用)

(Note: instruments and equipment with the same function can be used mutually)

6.3. 实验原理 Experimental Principle

分子排阻色谱法是根据待测组分的分子大小进行分离的一种液相色谱技术。分子排阻色谱法的分离原理为凝胶色谱柱的分子筛机制。色谱柱采用亲水键合硅胶为填充剂, 填充剂表面分布着不同尺寸的孔径, 抗体蛋白分子进入色谱柱后, 它们中的不同组分按其分子大小进行分离, 高分子聚合物, 单体分子, 低分子量物质依次洗脱出来。

Size-exclusion chromatography (SEC) is a liquid chromatography technique that separates constituents under test by molecular size. SEC separation principle is molecular sieve effect of gel chromatographic columns. A chromatographic column is filled with hydrophilic bonded silica gel whose surface is full of pores with different sizes. When the substances enter the column, different constituents wherein will be separated by molecular sizes. HMWS, Monomer molecules, and LMWS will be eluted sequentially.

6.4. 试剂配制 Reagents Preparation

(注: 6.4.1. - 6.4.2.试剂的配制均可以按照以下描述的倍数进行, 保证试剂成分含量不变即可。)

(Note: Reagents preparation described in 6.4.1.- 6.4.2 can be performed in multiples of all solutions, provided that component contents in each preparation remain unchanged.)

6.4.1. 氢氧化钠溶液 (4 mol/L)

Sodium hydroxide solution (4 mol/L)



标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程

Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

6.4.1.1. 称取80.0 g氢氧化钠, 加入500 mL超纯水, 混匀溶解。

Weigh 80.0 g of sodium hydroxide to 500 mL of ultrapure water, dissolve and mix well.

6.4.1.2. 溶液储存于试剂瓶中, 室温可储存3个月。

The solution may be stored for 3 months at room temperature.

6.4.2. 流动相 (含150 mmol/L氯化钠的100 mmol/L 磷酸盐缓冲液, pH 6.8±0.2)

Mobile phase (100 mmol/L sodium phosphate with 150 mmol/L sodium chloride, pH 6.8±0.2)

6.4.2.1. 称取6.1 g二水合磷酸二氢钠, 21.8 g十二水合磷酸氢二钠, 8.8 g氯化钠至950 mL超纯水, 溶解并混匀, 加入超纯水定容到1000 mL。测定pH应在6.8 (如pH不在6.8, 可用4 mol/L氢氧化钠溶液调节pH值至6.8±0.2)。0.45 μm滤膜过滤并超声10 min。

Weigh 6.1 g sodium dihydrogen phosphate dihydrate, 21.8 g of disodium hydrogen phosphate dodecahydrate and 8.8 g of sodium chloride to 950 mL of ultrapure water, and dissolve. Dilute to 1000 mL with ultrapure water. The pH should be measured at 6.8 (If the pH value is not 6.8, adjust the above solution to pH 6.8±0.2 with 4 mol/L sodium hydroxide). Filter the solution through a membrane filter of 0.45 μm and ultrasonicate for 10 min.

6.4.2.2. 溶液储存于试剂瓶中, 2-8°C可储存1个月。配制好的流动相放置时间超过3天, 需要重新用0.45 μm滤膜过滤后再使用。

The solution is preserved in a reagent bottle and may be stored for 1 month at 2-8°C. The mobile phase stored for more than 3 days needs to be filtered through a membrane filter of 0.45 μm before using.

6.5. 样品制备 Sample Preparation

空白对照、系统适应性样品、中控样品、研究性样品及其他样品均制备1份, 批检验及稳定性样品平行制备2份。(系统适应性样品采用现行参比品)。

(注: 制备时可通过温和吹吸或涡旋方式混匀, 吹吸次数不超过10次, 涡旋不超过30sec)。

Single preparation for blank, system suitability samples, in-process control samples, research samples and other samples. Duplicated preparation for batch release and stability samples. (The system suitability sample adopts the current reference sample).

(Note: Mixing can be operated by pipetting aspiration or vortexing gently during preparation, pipetting not more than 10 times, vortexing not more than 30 seconds.)

6.5.1. 系统适应性样品溶液的制备

Preparation of System Suitability Samples

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程**Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method**

用流动相将系统适应性样品稀释至1.0 mg/mL, 温和混匀, 单份制备。

Dilute appropriate volume of system suitability sample with mobile phase to 1.0 mg/mL, gentle mixing, single preparation.

6.5.2. 空白对照溶液的制备**Preparation of Blank**

将辅料溶液按照系统适应性样品溶液 (6.5.1.) 的方法制备成空白对照。

Dilute the excipient buffer according to the preparation of the system suitability samples (6.5.1.).

6.5.3. 供试品溶液的制备**Preparation of Samples**

本方法可检测0.3-2.0 mg/mL范围内的样品, 即进样量7.5-50 µg范围的样品。

This method can detect samples in the range of 0.3-2.0 mg/mL, which means sampling amounts of 7.5-50 µg.

6.5.3.1. 上游中控样品和PD步骤样品的前处理**Preparation of In-process control and PD samples for upstream**

样品参照标准操作规程 (BAD-TP-248) 进行纯化后, 再根据6.5.3.2~6.5.3.4 进行样品制备。

After the sample is purified according to the Standard Operating Procedure (BAD-TP-248), proceed with sample preparation based on sections 6.5.3.2 to 6.5.3.4.

6.5.3.2. 当供试品浓度大于1.0 mg/mL时, 用流动相将供试品稀释到1.0 mg/mL, 温和混匀, 进样体积25 µL。

If the concentration of sample is more than 1.0 mg/mL, dilute sample with mobile phase to 1.0 mg/mL, mix well, and inject 25 µL.

6.5.3.3. 当供试品浓度 ≥ 0.3 mg/mL, 且 ≤ 1.0 mg/mL时, 不需稀释, 进样体积为25 µL。

If the sample concentration was ≥ 0.3 mg/mL and ≤ 1.0 mg/mL, inject 25 µL.

6.5.3.4. 当供试品浓度 < 0.3 mg/mL, 且 ≥ 0.15 mg/mL时, 进样体积50 µL。

If the sample concentration was < 0.3 mg/mL and ≥ 0.15 mg/mL, inject 50 µL.

6.5.3.5. 当供试品浓度 < 0.15 mg/mL, 进样体积50 µL, 检测结果仅供参考。

If the sample concentration was < 0.15 mg/mL, inject 50 µL, and the result is for information only.

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程**Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method**

注: 如有必要, 可将系统适应性样品及供试品的浓度修约为整数, 用于计算。

(Note: If necessary, the concentration of the system suitability sample and the test sample can be rounded to the nearest integer for calculation).

6.5.4. 溶液稳定性 Stability of Prepared Sample Solution

上述制备完成的空白对照、系统适应性样品及供试品均可在6°C保存48 h。

The blank, system suitability sample and test samples prepared above can be stored at 6°C for 48 h.

6.6. 系统准备 System Preparation

系统冲洗和准备按照ASD安捷伦高效液相色谱仪操作规程 (BAD-EQ-003)。

System washing and set-up are performed in accordance with Agilent High-Performance Liquid Chromatography Operating Procedure (BAD-EQ-003).

6.7. 系统参数设定

Settings of System Parameters

6.7.1. 仪器: Agilent 1260

Instrument: Agilent 1260

6.7.2. 色谱柱: Nano BioCore SEC-300 (150×7.8 mm, 3 μm)

Column: Nano BioCore SEC-300 (150×7.8 mm, 3 μm)

6.7.3. 柱温: 室温 (不控制)

Column temperature: Uncontrolled

6.7.4. 样品室温度: 6°C

Autosampler temperature: 6°C

6.7.5. 检测器参数

Detector parameter

VWD检测器: 检测波长280 nm, 峰宽> 0.1 min (2s响应时间)

VWD Detector: wavelength 280 nm, peak width > 0.1 min (2s response time)

DAD检测器: 检测波长280 nm, 带宽4 nm, 峰宽> 0.1 min (2 s)

DAD Detector: wavelength 280 nm, bandwidth 4 nm, peak width > 0.1 min (2 s)

6.7.6. 泵参数设定: 流速0.800 mL/min; 停止时间10 min。

Pump parameter setting: flow rate 0.800 mL /min; Stop time 10 minutes.

6.8. 系统平衡 System Equilibration

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程
Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

6.8.1. 可根据实际的情况判断是否需要系统进行平衡。

Equilibrate the system if necessary.

6.8.2. 若进行系统平衡, 可进样空白对照 (6.5.2.) 或系统适应性样品 (6.5.1.)。

To equilibrate the system, the blank (6.5.2.) or system suitability sample (6.5.1.) could be injected.

6.9. 进样序列

Injection Sequence

6.9.1. 中控样品、研究性样品和其他样品进样序列

Sequences of in-process control samples, research samples and other samples.

序列顺序 Sequence Order	类型 Type	样品名称 Sample Name	进样次数 Number of Injection
1	系统适应性 System suitability	空白对照 Blank	1
2		系统适应性样品 System suitability samples	1
3	供试品 (1-N) Test sample 1-N	供试品 1 Test sample 1	1
4		供试品 2 Test sample 2	1
.....			
.....		供试品 N Test sample N	1
.....	系统适应性 System suitability	系统适应性样品 System suitability samples	1

注: 序列开始进一针空白对照和一针系统适应性样品, 最多进样10针供试品需进样一针系统适应性样品, 最后一个供试品后再进样一针系统适应性样品。

Note: Start the sequence with one injection of blank and system suitability sample, thereafter one injection of the system suitability sample should follow no more than 10 injections of test samples and the last sample injection.

6.9.2. 批检验及稳定性样品进样序列

Sequences of batch release and stability tests.

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程
Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

序列顺序 Sequence Order	类型 Type	样品名称 Sample Name	进样次数 Number of Injection
1	系统适应性 System suitability	空白对照 Blank	1
2		系统适应性样品 System suitability samples	2
3	供试品 (1-N) Test sample 1-N	供试品 1-1 Test sample 1-1	1
4		供试品 1-2 Test sample 1-2	1
.....			
.....		供试品 N-1 Test sample N-1	1
.....		供试品 N-2 Test sample N-2	1
.....	系统适应性 System suitability	系统适应性样品 System suitability samples	1

注: 序列开始进一针空白对照和两针系统适应性样品, 最多进样10针供试品需进样一针系统适应性样品, 最后一个供试品后再进样一针系统适应性样品。

Note: Start the sequence with one injection of blank and two injections of system suitability samples, thereafter one injection of the system suitability sample should follow no more than 10 injections of test samples and the last sample injection.

6.10. 系统清洗及色谱柱保存

System Washing and Column Storage

6.10.1. 色谱柱保存按照理化和E&L色谱柱使用标准操作规程 (BAD-GE-014) 进行操作, 参照纳谱5 μm 色谱柱的保存维护方法。

The column is preserved in accordance with The Operational Procedure of Chromatographic Column for Physicochemical and E&L (BAD-GE-014), referring to the preservation and maintenance method of Nano 5 μm column.

6.10.2. 系统清洗按照安捷伦高效液相色谱仪操作规程 (BAD-EQ-003) 进行操作。

The system is washed in accordance with Agilent High Performance Liquid Chromatography Operating Procedure (BAD-EQ-003).

6.11. 数据分析 Data Analysis

6.11.1. 系统适应性样品、供试品色谱图与空白对照色谱图进行对比, 调整积分事件, 确

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程**Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method**

保在目标峰出峰位置所有高分子聚合物、单体和低分子量物质的峰都得到积分, 空白辅料峰位置不积分。

Compare chromatograms of the system suitability sample, test sample, and blank control, adjust integration events to ensure that all HMWS、Monomer and LMWS have been integrated except those peaks presented in the blank. Peak of blank excipient is not integrated.

6.11.2. 积分参数设置

Integral parameter setting

积分参数 Integration Parameter	初始值 Initial value	建议调整范围 Recommend adjustment range
Integration Algorithm	Apex Track (建议) Apex Track (Recommendation)	
Integration Start	3.5 min	
Integration End	6.6 min	
Peak Width	36	20-40
Threshold	0	0-5

*如上述参数范围不适用, 可根据实际情况, 按照BAD-GE-053 ASD理化色谱峰的鉴别和积分处理步骤进行合理积分, 如添加Minimum Height和Detected Shoulder积分事件等。

*Note: If the above parameters are not applicable, reasonable integration can be carried out based on the actual situation, according to the BAD-GE-053 ASD Chromatographic Peak Identification and Integration Procedures, such as adding Minimum Height and Detected Shoulder etc.

6.11.3. 色谱峰命名 Naming of chromatographic peaks

各组分按分子量大小先后洗脱出来, 洗脱顺序依次是高分子聚合物、单体组分、低分子量物质。色谱峰面积最大是单体组分, 单体之前是高分子聚合物, 单体之后是低分子量物质。

When the substances enter the column, different constituents wherein will be separated by molecular sizes. HMWS, monomer molecules and LMWS will be eluted sequentially. The largest chromatographic peak is the Monomer component, which is preceded by HMWS and followed by LMWS.

6.11.4. 方法保存 Saving method

保存积分方法并命名。

Name and save the processing method.

**标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程****Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method**

6.11.5. 结果计算 Result calculation

按照面积归一化法计算色谱峰峰面积, 计算公式详见6.15.

Peak areas% are calculated using area normalization method, while the calculation formulas are detailed as 6.15.

6.12. 图谱处理 Chromatogram Processing

6.12.1. 批检验及稳定性样品、中控样品和特殊研究性样品, 保存并打印样品组方法、图谱报告、积分方法和仪器方法。

The sample set method, chromatograms report, processing method and instrument method should be saved and printed for batch release and stability samples, in-process control samples, special research samples.

6.12.2. 其他样品保存并打印样品组方法和图谱报告。

The sample set method, chromatograms report should be saved and printed for other samples.

6.13. 系统适应性考察 System Suitability

6.13.1. 空白对照溶液在目标峰出峰位置无干扰峰, 或空白对照色谱图于目标峰出峰位置的色谱峰面积不大于第一针系统适应性样品总峰面积的1.0%。

The blank appearing at the location of target peak must have no interference peak or the interference peak area of blank appearing at the location of target peak is not more than 1.0% of the total peak area of the first injection system suitability sample.

6.13.2. 同一序列中的系统适应性样品单体峰的理论塔板数 (EP) ≥ 4000 。

The theoretical plate count of the monomer peak of system suitability samples in the same sequence must be no less than 4000.

6.13.3. 同一序列的系统适应性样品单体峰面积%的RSD $\leq 2.0\%$, 如只有两针系统适应性样品, 单体峰峰面积%的RD $\leq 2.0\%$ 。

The RSD of monomer% of all system suitability samples in the same sequence must be no more than 2.0%, and the RD of monomer% must be no more than 2.0% if there are only two-needle system adaptable samples.

6.14. 结果报告 Result Reporting

6.14.1. 报告内容 Content of report

本方法的定量限为0.02%, 检测限为0.01%, 若峰面积%的检测结果低于0.02%但不低于0.01%时, 则汇报“<0.02%”, 即低于定量限, 检测结果仅供参考; 若峰面积%的检测结果低于0.01%时, 则报告“ND”(ND:Not detected)。

The LOQ of the reduced method is 0.02% and the LOD is 0.01%. If the area percentage



标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程

Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

is less than 0.02% but not less than 0.01%, reporting '<0.02%' for reference. If the area percentage is less than 0.01%, reporting 'ND'.

6.14.1.1. 中控、研究性及其他样品: 报告高分子聚合物峰面积%、单体峰面积%和低分子量物质峰面积%, 数据精确到0.1%, 若出现0.0%或0.00%结果, 则保留一位有效数字。

For in-process control samples, research samples and other samples, reporting the HMWS%, Monomer% and LMWS% of each injection. Data is accurate to 0.1%. If 0.0% or 0.00% occur after the result has been modified, the report result is allowed to retain one significant digit.

6.14.1.2. 批检验及稳定性样品: 报告两针供试品高分子聚合物峰面积%平均值、单体峰面积%平均值和低分子量物质峰面积%平均值, 数据精确到0.1%, 如出现0.0%或0.00%结果, 则保留一位有效数字。

For batch release and stability samples, reporting the mean of HMWS%, Monomer% and LMWS%. Data is accurate to 0.1%. If 0.0 or 0.00 occur after the result has been modified, the report result is allowed to retain one significant digit.

6.15. 计算公式

Calculation Formulas

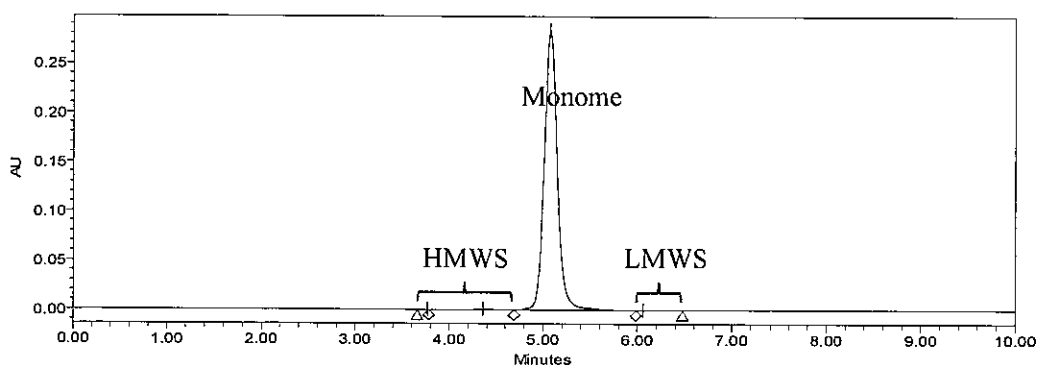
序号 No.	公式 Formulas	公式注解 Explanation
1	高分子聚合物含量% = $\sum$ 高分子聚合物峰面积% $\text{HMWS content\%} = \sum \text{HMWS peak area\%}$ $= \sum \frac{A_{\text{HMWS}}}{\text{Total Peak Area}} \times 100\%$	A_{HMWS} : 高分子聚合物峰面积 A_{HMWS} : HMWS peak area
2	单体纯度% = 单体峰面积% $\text{Monomer content\%} = \sum \text{Monomer peak area\%}$ $= \frac{A_{\text{monomer}}}{\text{Total Peak Area}} \times 100\%$	A_{monomer} : 单体峰面积 A_{monomer} : monomer peak area
3	低分子量物质含量% = 低分子量物质峰面积% $\text{LMWS content\%} = \sum \text{LMWS peak area\%}$ $= \sum \frac{A_{\text{LMWS}}}{\text{Total Peak Area}} \times 100\%$	A_{LMWS} : 低分子量物质峰面积 A_{LMWS} : LMWS peak area

标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程

Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

序号 No.	公式 Formulas	公式注解 Explanation
4	$\bar{X} = \frac{\sum_{i=1}^n X}{n}$	$\bar{X}$: 峰面积百分比平均值 $\bar{X}$: Mean peak area% X : 峰面积百分比 X : Peak area% n : 峰面积百分比的数目 n : Number of peak areas%
5	$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$	S : 峰面积百分比标准偏差, 也可用 SD 表示 S : Standard deviation of peak area%, or denoted by SD
6	$RSD = \frac{S}{\bar{X}} \times 100\%$	RSD: 峰面积百分比相对标准偏差 RSD: Relative standard deviation of peak area%
7	$RD = \frac{(A - B)}{(A + B)/2} \times 100\%$	RD: 相对差异百分比 RD: Relative Difference A: 样品 A 的检测值 (较大值) A: Detected value of sample A (larger value) B: 样品 B 的检测值 (较小值) B: Detected value of sample B (smaller value)

6.16. 典型图谱 Typical Chromatograms



标题: HLX49-Ab 分子排阻色谱检测方法标准操作规程
Title: Standard Operating Procedure of HLX49-Ab Size Exclusion Chromatography Method

图1 HLX49-Ab代表性样品典型色谱图（全图）

Figure 1 Typical chromatograms of HLX49-Ab representative sample (Full-scale)

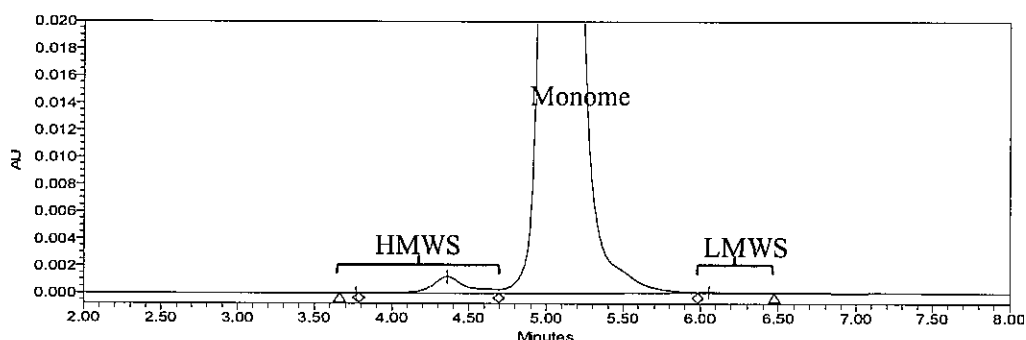


图2 HLX49-Ab代表性样品典型色谱图（细节图）

Figure 2 Typical chromatograms of HLX49-Ab representative sample (Expanded)

7. 附件 ATTACHMENTS

7.1. HLX49-TP-002A, HLX49-Ab分子排阻色谱法检测记录

HLX49-TP-002A, Test Record of HLX49-Ab Size Exclusion Chromatography Method.

8. 版本历史 REVISION HISTORY

版本 Version	修订内容 Change
01	新文件 New document
02	- 新增6.5.3本方法的检测样品范围。 Add the detection sample range of this method in 6.5.3. - 修改6.11.2积分参数设置参考值。 Modify reference value for integration parameter settings in 6.11.2. - 根据方法LOD和LOQ, 更新6.14结果汇报的要求。 Update the 6.14 result reporting requirements according to the LOD and LOQ. - 修改6.13.2中单体峰的理论塔板数的考察。 Modify the requirement for theoretical plate count of the monomer peak in 6.13.2. - 根据正文更新附件相应内容。 According to the main text, update the corresponding contents of the attachment.

附件编号 Attachment No. HLX49-TP-002A	版本 Ver. 02	Page 1 of 4
标题: HLX49-Ab 分子排阻色谱法检测记录 Title: Test Record of HLX49-Ab Size Exclusion Chromatography Method		

注: 英文部分只做参考, 不需填写。

Note: English for reference only, do not fill it.

检验日期: Test date		检验记录编号: Test record number
序号 No.	ASD样品号 ASD Sample Number	样品名称 Sample Name
1		
2		
3		
4		
5		

1. 溶液信息 Solution Information

样品类型 Sample type	厂家/货号 Manufacturer and Cat. No.	批号/配制编号 Batch No. /Preparation No.	有效期至 Expiration Date
系统适应性样品 System Suitability Sample	Henlius		
辅料溶液 Excipient solution	实验室配制或分装 Laboratory preparation or sub package		
流动相 Mobile Phase	实验室配制 Laboratory preparation		

2. 样品制备 Sample Preparation

2.1. 系统适应性样品、空白对照溶液制备 System Suitability Samples and Blank Preparation

样品类型 Sample Type	浓度 Concentration (mg/mL)	样品加入量 Sample Volume (μL)	流动相加入量 Mobile Phase Volume (μL)	制备方式 Preparation Method
空白对照溶液 Blank	N/A			单份制备 Prepared in single
系统适应性样品 System Suitability Sample				单份制备 Prepared in single

检验人 operator by : _____ 日期 date: _____

CONFIDENTIAL

2.2. 样品制备 Test Sample Preparation

序号 No.	浓度 Concentration (mg/mL)	样品加入量 Sample Volume (μL)	流动相加入量 Mobile Phase Volume (μL)	制备方式 Preparation Method
1				☐ 单份制备 Prepare in single ☐ 双份制备 Prepare in duplicate
2				☐ 单份制备 Prepare in single ☐ 双份制备 Prepare in duplicate
3				☐ 单份制备 Prepare in single ☐ 双份制备 Prepare in duplicate
4				☐ 单份制备 Prepare in single ☐ 双份制备 Prepare in duplicate
5				☐ 单份制备 Prepare in single ☐ 双份制备 Prepare in duplicate

3. 仪器和色谱柱 Instrument and Column

仪器/色谱柱 Instrument/Column	厂家/型号/参数 Manufacturer/Cat. No./ Parameter			设备编号/序列号 Equipment Number/Serial No.
HPLC	Agilent 1260			
	柱温 Column temperature (°C)	流速 Flow rate (mL/min)	检测波长 Wavelength (nm)	
色谱柱 Colum	Nano BioCore SEC-300 (150×7.8 mm, 3 μm)			

4. 结果 Results

4.1. 结果保存 Save Path

Project Name	
Sample Set Name	
Acq. Method Name	
Processing Method	

4.2. 系统适应性 System Suitability

4.2.1. 空白对照 Blank

检验人 operator by : _____ 日期 date: _____

标题: HLX49-Ab 分子排阻色谱法检测记录

Title: Test Record of HLX49-Ab Size Exclusion Chromatography Method

考察内容 System Suitability Test	结果 Result	结论 Conclusion
空白对照色谱图在目标峰出峰位置无干扰, 或空白对照色谱图于目标峰出峰位置的色谱峰面积不大于第一针系统适应性样品总峰面积的 1.0%。 The blank appearing at the location must have no interference peak or the interference peak area is not greater than 1.0% of the total peak area of the first injection of system suitability sample.	☐ 无干扰峰 no interference peak ☐ 有干扰峰 interference peak, Area%: _____	☐ 通过 Pass ☐ 不通过 Fail

4.2.2. 系统适应性样品结果(峰面积%报告数据精确到 0.1%, 如出现 0.0%或 0.00%结果, 则保留一位有效数字)

System Suitability results (Peak area% data accurate to 0.1%. If 0.0% or 0.00% occur after the result has been modified, the report result is allowed to retain one significant digit)

样品名称 Sample name	高分子聚合物 峰面积% HMWS%	单体峰面积% Monomer%	低分子量物质 峰面积% LMWS%	单体峰 理论塔板数 EP of Monomer
☐ RSD ☐ RD	N/A		N/A	

4.2.3. 系统适应性样品考察 System Suitability

考察内容 Contents of investigation	结论 Conclusion
同一序列中系统适应性样品单体峰峰面积%的 RSD $\leq$ 2.0%, 如只有两针系统适应性样品, 单体峰面积%的 RD $\leq$ 2.0% The RSD of monomer% of all system suitability samples in the same sequence must be no more than 2.0%, and the RD of monomer% must be no more than 2.0% if there are only two-needle system adaptable samples.	☐ 通过 Pass ☐ 不通过 Fail
每针系统适应性样品单体峰理论塔板数 (EP) $\geq$ 4000 The theoretical plate count (EP) of the monomer peak of system suitability samples in the same sequence must be no less than 4000.	☐ 通过 Pass ☐ 不通过 Fail

4.3. 结果: 报告数据精确到 0.1% (峰面积%报告数据精确到 0.1%, 如出现 0.0%或 0.00%结果, 则保留一位有效数字)。

Results: Report data is accurate to 0.1% (Peak area% data accurate to 0.1 %. If 0.0% or 0.00% occur after the result has been modified, the report result is allowed to retain one significant digit).

检验人 operator by : _____ 日期 date: _____

CONFIDENTIAL

ASD 样品号 ASD Sample Number	样品名称 Sample Name	高分子聚合物 峰面积 (%) HMWS%	单体峰面积 (%) Monomer%	低分子量物质 峰面积 (%) LMWS%

注: 本方法的定量限为 0.02%, 检测限为 0.01%, 若高分子聚合物峰面积%和低分子量物质峰面积%的检测结果低于 0.02%但不低于 0.01%时, 则报告 “<0.02%”, 即低于定量限, 检测结果仅供参考; 若高分子聚合物峰面积%和低分子量物质峰面积%的检测结果低于 0.01%时, 则报告 “ND” (ND: Not detected)。

Note: The LOQ of the reduced method is 0.02% and the LOD is 0.01%. If the HMWS% and the LMWS% are less than 0.02% but not less than 0.01%, reporting ‘<0.02%’ for reference. If the HMWS% and the LMWS% are less than 0.01%, reporting ‘ND’.

5. 附件 Attachment

序列表附后见附件_____, 共_____页。

Sequence table refer to attachment_____, total pages_____.

原始图谱附后见附件_____, 共_____页。

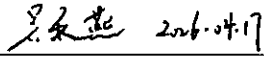
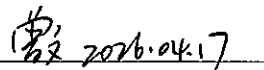
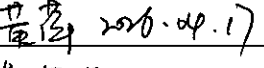
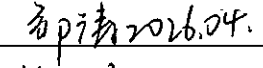
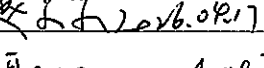
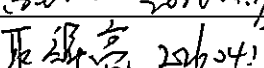
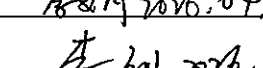
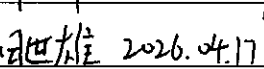
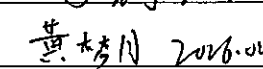
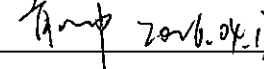
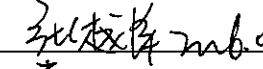
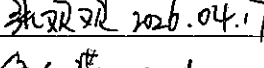
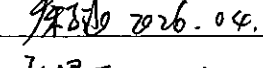
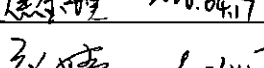
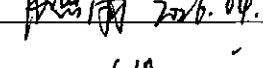
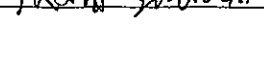
Original chromatograms refer to attachment_____, total pages_____.

6. 备注 Note

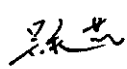
检验人 operator by: _____ 日期 date: _____

复核人 Reviewed by: _____ 日期 date: _____

附件编号 Attachment No. PD-GE-014B	版本 Ver. 03	Page 1 of 1
标题：研发部门培训记录 Title: Training Record of R&D Department		

培训主题 Training Subject	HLX49-Ab 分子排阻色谱法检测方法标准操作规程 HLX49-TP-002, Ver.02		
培训人签名/日期 Trainer /Date	 2026.04.17	培训地点 Training Location	松江一厂
培训日期 Training Date	2026.04.17	培训课时 Training Hours	0.5h
培训类型 Training Type	☒ 文件培训 Document training ☐ GxP ☐ 外部培训 External training ☐ 其他 Other: _____		
培训方式 Training Method	☒ 阅读理解 Reading Comprehension ☐ 课堂培训 Classroom Training ☐ 操作培训 Operation Training ☐ 其他 Other: _____		
评估方式* Assessment Type	☐ 书面测试 Written Test ☒ 口头问答 Oral Q&A ☐ 电子测试 Electronic Test ☐ 操作 Operation ☐ 无需评估 No Assessment ☐ 其他 Other: _____		
考核标准 Assessment Criteria	测试正确率 100%通过为合格。		
部门 Department	受训人签名/日期 Trainee's Signature/Date	部门 Department	受训人签名/日期 Trainee's Signature/Date
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	ASD	 2026.04.17
ASD	 2026.04.17	N/A	N/A

附件编号 Attachment No. PD-GE-014B	版本 Ver. 03	Page 1 of 1
标题：研发部门培训记录 Title: Training Record of R&D Department		

培训是否合格 Pass or not	☒ 是 Yes (以上所有受训人均通过培训 All the above trainees pass the training) ☐ 否 No (请于下方空白处列出未达到考核标准者, 除下方所列人员, 其他受训人均通过培训 Please list in the space below those who do not meet the assessment criteria, except for those listed below, all other trainees pass the training) 评估人/日期:  Evaluated By/Date: 2016.04.17
备注 Notes	N/A

*口头问答将相应的问题和答案附后, 电子测试将学习记录附后, 操作将实验记录或其复印件附后。
 Oral Q&A attaches the questions and answers. Electronic test attaches learning records. Operation attaches experimental records or their copies.